

SET-3



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Department of Computer Science & Engineering

Ist Lab Internal - Set 3

Subject Name: Software Engineering Subject Code: 23CC505PC

Year and Semester: III / I Branch/Section: CSE – A

Faculty: K Yasaswini Lab: FS2

In today's competitive job market, companies receive thousands of job applications, and managing recruitment manually becomes time-consuming and error-prone. To streamline this process, your development team has been tasked with building a **web-based Online Recruitment System**. This system should allow job seekers to apply for positions, while employers can post job openings, review applications, and schedule interviews efficiently.

1. **Clone the Online Recruitment System repository to your eclipse software from the URL below and execute the project by changing the required changes in the POM.xml file and make it a successful build. [40 Marks]**

<https://github.com/ssvkotamraju/ORS.git>

- a. What steps would you follow to clone the repository, import it correctly into Eclipse, and verify that the pom.xml file is recognized?
- b. How would you identify the problematic dependencies, modify pom.xml, update the Maven project, and verify that the project builds successfully?
- c. How would you identify the dependency conflict and modify pom.xml to use a compatible version?
- d. What changes would you make in pom.xml and Eclipse to resolve the Java version compatibility issue and successfully execute the project?
- e. Explain the steps you would follow to troubleshoot the issue, clean the project, refresh Maven dependencies, and confirm a successful build.
- f. How would you diagnose the problem and configure the required server/port settings so that the Online Recruitment System can be accessed through a browser?
- g. How would you identify the application using port 8080 and configure the ORS application to run on another available port?

- h. How would you identify the Maven compiler configuration causing the problem and modify the pom.xml to use a Java version supported by your installed JDK?
- i. How would you determine whether the problem is caused by an incorrect dependency declaration, repository configuration, network connectivity, or Maven's local cache?
- j. What Maven commands and configuration checks would you perform to identify differences in the JDK, Maven version, local repository, and project dependencies?

2. Working with Version Control (GitHub, GitLab, etc.) [40 Marks]

Your team is developing an **Online Recruitment System**. You are working on the job application form (apply.jsp). You will collaborate with teammates by pulling remote changes, rebasing your work, resolving merge conflicts, and pushing the final version to GitHub using SSH key.

- a. How would you clone the GitHub repository, create a working branch, and verify that your local repository is correctly connected to the remote repository?
- b. Before pushing your changes, how would you retrieve the latest remote changes and determine whether your branch is behind the remote branch?
- c. What Git commands would you use to update your local branch and rebase your changes onto the latest remote branch?
- d. How would you identify the conflict, decide which changes to keep, edit the file, stage it, and continue the rebase?
- e. How would you systematically resolve all the conflicts and verify that the rebase completed successfully?
- f. How would you investigate the commit history and restore the missing changes without unnecessarily losing your own work?
- g. How would you cancel the rebase safely and return your branch to its previous state?
- h. What steps would you take to configure an SSH key, add the public key to GitHub, test the SSH connection, and retry the push?
- i. How would you verify that the correct version of apply.jsp is present on GitHub and that your commits contain only the intended changes?
- j. What type of push would be appropriate after a rebase, and what precautions should you take before using it on a shared branch?

3. Run the Project with Docker [20 Marks]

- i. Create a Dockerfile and build a Docker image named ORS to push it to DockerHUB.
 - ii. Run the image in a container and expose port 8080. Check if the app runs at <http://localhost:8080>.
- a) What information would you need before creating the Dockerfile, and what major instructions would you include to build a Docker image capable of running the ORS application?
 - b) How would you troubleshoot the Dockerfile, project structure, build artifact, and Docker build context?
 - c) How would you run the container so that container port 8080 is correctly mapped to host port 8080?

d) What changes could you make to the Dockerfile and Docker build process to reduce the image size and improve build efficiency?

e) Explain how you would tag the image appropriately, authenticate with Docker Hub, push the image, and verify that it is available in the repository.